DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ARTIFICIAL INTELLGENCE & MACHINE LEARNING

Alternate Assessment Tool

Course Title: INTERNET OF THINGS AND 5G
Course Code: 22AI642
Semester: VI

Submitted by

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Under the Guidance of
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2025-26

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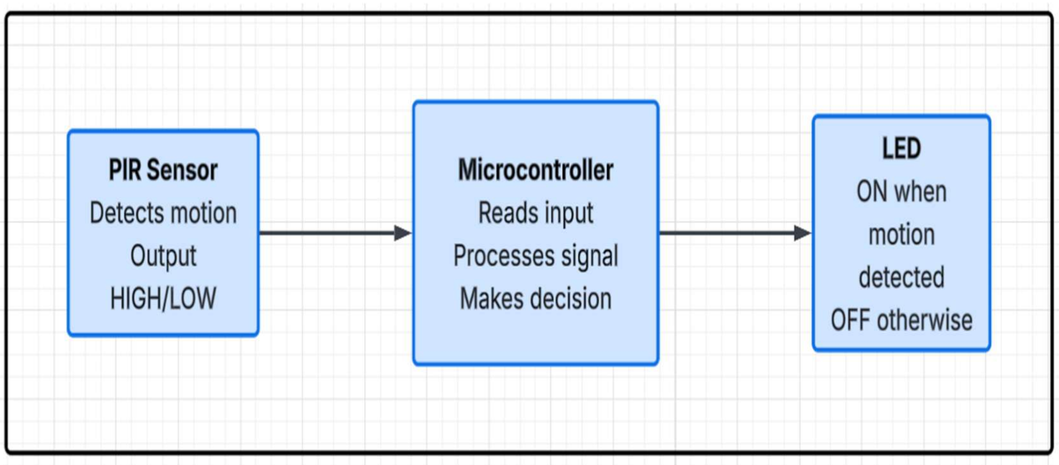
Activity 1: TinkerCAD Aurdino

<table>
<tr><td>Problem Statement</td><td colspan="5">In modern environments such as residential buildings, offices, and restricted areas, continuous monitoring of human movement is essential for ensuring safety and security. Conventional surveillance systems are often complex, expensive, and require constant human supervision. Therefore, there is a need for a simple, cost-effective, and automated solution that can detect motion in real time and provide immediate alerts. This project aims to design a motion sensor-based system using Arduino that can efficiently identify human movement and trigger a visual indication.</td></tr>
<tr><td>AAT
MARKS</td><td>Design (5)</td><td>Execution (5)</td><td>Result (5)</td><td>Innovation/ Extra Features (5)</td><td>Total</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>TOOLS USED</td><td colspan="5">• TinkerCAD
• Arduino Uno
• PIR Motion Sensor
• LED</td></tr>
<tr><td>MAP SDG</td><td colspan="5">SDG 3: Good Health and Well-being
The system enhances safety by detecting human movement and providing timely alerts to prevent potential risks.
SDG 11: Sustainable Cities and Communities
Supports the development of safer environments through automated monitoring and basic surveillance systems.
SDG 9: Industry, Innovation and Infrastructure
Demonstrates the use of IoT and embedded systems to develop a simple and efficient motion detection solution.</td></tr>
<tr><td>MAP COURSE OUTCOME of IOT</td><td colspan="5"><table><tr><th>CO</th><th>Mapping</th><th>Justification</th></tr></table></td></tr>
</table>

	CO1	Yes	Demonstrates perception layer of IoT architecture using PIR sensor to detect motion, microcontroller processes the signal, and LED acts as output.
	CO2	Yes	Uses Arduino C programming, digital input reading, and basic signal processing to detect motion and trigger output.
	CO3	No	Data analytics is not applied as the system focuses only on real-time motion detection and response.
	CO4	Yes	Integrates PIR sensor (sensing) and LED (actuation) to implement a complete sense-process-act IoT system.
MAP PROGRAM OUTCOME	**PO**	**Mapping**	**Justification**
	PO1	Yes	Applied knowledge of electronics (sensors), basic science, and programming for detect motion.
	PO2	Yes	Identified and analysed the real-world problem of security and motion detection for automated response.
	PO3	Yes	Designed a simple motion detection system for automation and security applications.
	PO5	Yes	Used modern tools such as Arduino IDE and simulation platforms for system design and implementation.
	PO6	Yes	Addresses societal safety by detecting unauthorized movement and triggering alerts.
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	**PSO**	**Mapping**	**Justification**
	PSO 1	Yes	Demonstrates programming and logical implementation of motion detection using sensor input and conditional statements.

<table>
<tr><td></td><td>
<table>
<tr><td>PSO 2</td><td>Yes</td><td>Identifies and solves the real-world problem of intrusion detection using automated sensing systems</td></tr>
<tr><td>PSO 3</td><td>Yes</td><td>Utilizes simulation tools like Tinkercad or embedded platforms to design, test, and deploy motion detection system</td></tr>
</table>
</td></tr>
<tr><td>URL Link</td><td>https://www.tinkercad.com/things/1F2gf7Qakvg-munagala-chandu</td></tr>
<tr><td>SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model</td><td>


</td></tr>
<tr><td>COURSE COORDINATOR</td><td>Dr ARUNA M G</td></tr>
</table>

Details
Abstract : The Motion Detection System is an IoT-based project developed using Arduino Uno and a PIR motion sensor. The system detects human movement and provides a visual alert using an LED. When motion is detected, the LED turns ON, and when no motion is present, the LED remains OFF. This project demonstrates basic sensor interfacing and real-time response, making it useful for simple security and monitoring applications.

Use Case:

This system can be used in homes, offices, and restricted areas for basic security and monitoring. It helps in detecting unauthorized movement and provides immediate visual indication, making it useful for surveillance and automation systems.

Introduction:

Security and monitoring are essential in modern environments to ensure safety. Traditional surveillance systems can be expensive and complex. A simple motion detection system using a PIR sensor provides an efficient and low-cost alternative. The PIR sensor detects infrared radiation changes caused by human movement and sends signals to the Arduino, which then triggers an alert. This project demonstrates how embedded systems can be used for real-time monitoring.

Objective:

- To detect human motion using a PIR sensor
- To interface and process sensor data using Arduino Uno
- To trigger a visual alert (LED) upon motion detection
- To monitor motion status through serial communication
- To develop a simple and cost-effective system

General procedure

- Connect PIR sensor to Arduino (VCC, GND, OUT)
- Connect LED to output pin
- Initialize input and output pins in code
- Continuously read motion data from PIR sensor
- Check motion status
- Turn LED ON or OFF accordingly

Step to Execute:

- Open Tinkercad simulation platform

- Add Arduino Uno, PIR sensor, and LED
- Connect components properly
- Write or paste the Arduino code
- Click on "Start Simulation"
- Open Serial Monitor
- Trigger motion using PIR sensor toggle
- Observe LED and output messages

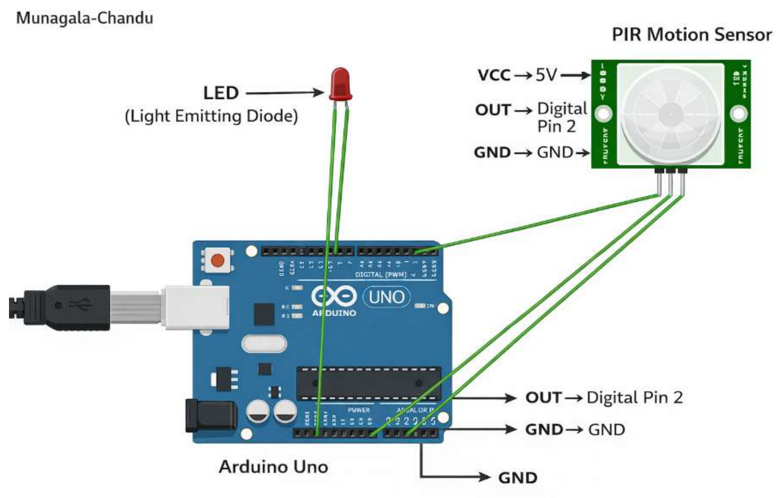
Source Code :

```
int pir = 2;

int led = 7;

void setup() {

  pinMode(pir, INPUT);

  pinMode(led, OUTPUT);

  Serial.begin(9600); }

void loop() {

  int motion = digitalRead(pir);

  if (motion == HIGH) {

    digitalWrite(led, HIGH);

    Serial.println("Motion Detected!");

  } else {digitalWrite(led, LOW);

    Serial.println("No Motion");}

  delay(500);}
```

Output:

```
No Motion
No Motion
No Motion
No Motion
No Motion
No Motion
No Motion
No Motion
No Motion
No Motion
Motion Detected!
Motion Detected!
Motion Detected!
Motion Detected!
Motion Detected!
Motion Detected!
Motion Detected!
Motion Detected!
Motion Detected!
Motion Detected!
No Motion
No Motion
```

Photos/ Screenshot/Snapshot :



Outcome:

The Motion Detection System successfully detects human movement using a PIR sensor and triggers an LED alert. The system demonstrates real-time monitoring and response, highlighting the application of IoT in basic security systems.

Activity 2: Wokwi ESP32

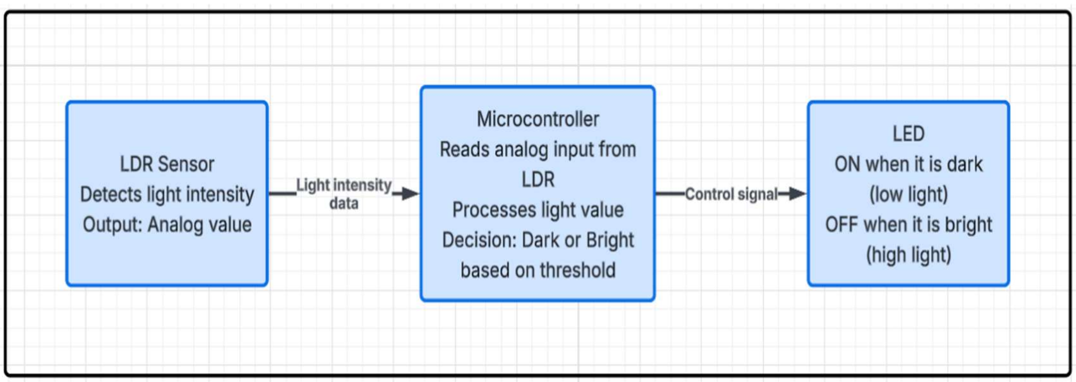
<table>
<tr><td>Problem Statement</td><td colspan="5">In many environments, efficient energy usage is essential, especially in lighting systems. Manual control of lights can lead to unnecessary power consumption during daylight conditions. Therefore, an automated system is required to monitor ambient light intensity and control lighting accordingly.</td></tr>
<tr><td>AAT MARKS</td><td>Design (5)</td><td>Execution (5)</td><td>Result (5)</td><td>Innovation/ Extra Features (5)</td><td>Total</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>TOOLS USED</td><td colspan="5">• Wokwi
• ESP32
• LDR
• LED</td></tr>
<tr><td>MAP SDG</td><td colspan="5">• SDG 7: Affordable and Clean Energy
Promotes efficient energy usage by automatically controlling lighting based on ambient conditions.
• SDG 11: Sustainable Cities and Communities
Supports smart and energy-efficient environments through automated lighting systems.
• SDG 9: Industry, Innovation and Infrastructure
Demonstrates IoT-based automation using embedded systems for real-world applications.</td></tr>
<tr><td>MAP COURSE OUTCOME of IOT/ BFE</td><td colspan="5"><table><tr><th>CO</th><th>Mapping</th><th>Justification</th></tr><tr><td>CO 1</td><td>Yes</td><td>Demonstrates IoT perception layer using LDR sensor to detect light intensity, with microcontroller processing and controlling output</td></tr></table></td></tr>
</table>

	CO 2	Yes	Uses Arduino C programming, analog data acquisition, and threshold-based decision logic for light detectionconversion, and communication protocols for real-time monitoring
	CO 3	No	Data analytics is not applied as the system performs real-time detection without storing or analyzing data
	CO 4	Yes	Integrates LDR sensor (sensing) and LED (actuation) to implement a complete sense-process-act IoT system
MAP PROGRAM OUTCOME	**PO**	**Mapping**	**Justification**
	PO1	Yes	Applied knowledge of sensors, electronics, and programming to measure light intensity and control LED
	PO2	Yes	Identified and analyzed real-world problems such as automatic lighting based on environmental conditions
	PO3	Yes	Designed a system to automatically control lighting based on ambient light levels
	PO5	Yes	Used modern tools such as Arduino IDE and simulation platforms for system implementation
	PO6	Yes	Contributes to energy efficiency by enabling automatic light control systems

<table>
<tr><td>MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT</td><td>

PSO	Mapping	Justification
PSO 1	Yes	Demonstrates programming and logical implementation of light detection using sensor input and threshold conditions
PSO 2	Yes	Solves real-world problems like automatic lighting systems for homes and streets
PSO 3	Yes	Utilizes simulation tools like Wokwi to design, test, and validate the light detection system

</td></tr>
<tr><td>URL Link</td><td>https://wokwi.com/projects/461373913268265985</td></tr>
<tr><td>SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model</td><td>



</td></tr>
<tr><td>COURSE COORDINATOR</td><td>Dr ARUNA M G</td></tr>
</table>

Details
Abstract : The Automatic Light Control System is an IoT-based project developed using ESP32 and an LDR sensor. The system monitors ambient light intensity and automatically controls an LED. When the surrounding environment is dark, the LED turns ON, and when it is bright, the LED turns OFF. The system also displays light intensity values in the serial monitor, demonstrating real-time monitoring and automation.

Use Case:

This system can be used in automatic street lighting, smart homes, and outdoor lighting systems where lights need to operate based on surrounding light conditions. It helps in reducing energy consumption by ensuring that lights are turned ON only when it is dark and turned OFF during daylight.

Introduction:

Efficient energy usage is important in modern lighting systems. Manual control of lights can lead to unnecessary power consumption. This project uses an LDR sensor and ESP32 to automatically control lighting based on ambient conditions. It demonstrates a simple and energy-efficient IoT solution.

Objective:

- To detect ambient light intensity using an LDR sensor
- To interface LDR with ESP32
- To control LED based on light conditions
- To display sensor values using serial monitor
- To develop a simple automated lighting system

General procedure

- Connect LDR module to ESP32 (VCC, GND, AO)
- Connect LED to output pin
- Initialize serial communication
- Read analog values from LDR
- Compare with threshold value
- Control LED based on light intensity

Step to Execute:

- Open Wokwi simulation platform
- Add ESP32, LDR module, and LED
- Connect components properly
- Paste the code in the editor
- Click on "Start Simulation"
- Open Serial Monitor
- Adjust light intensity and observe LED behavior

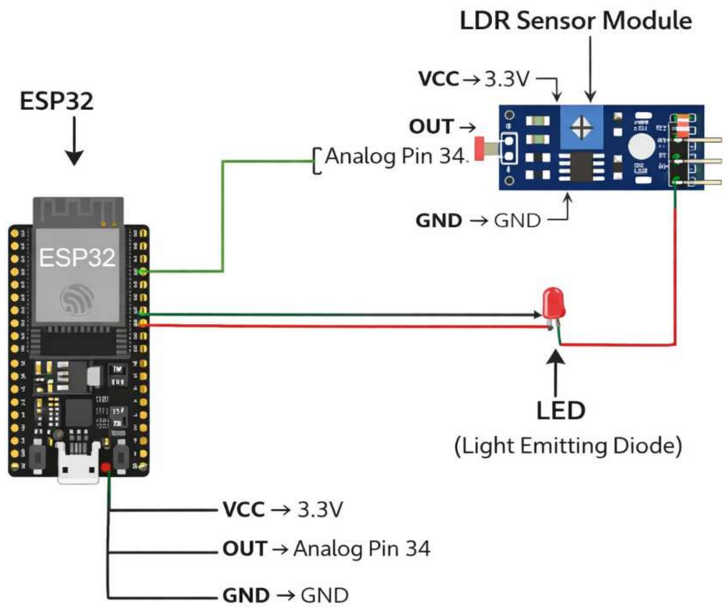
Source Code :

```
int ldr = 34;
int led = 2;

void setup() {
  pinMode(led, OUTPUT);
  Serial.begin(115200);
}

void loop() {
  int light = analogRead(ldr);

  Serial.print("Light: ");
  Serial.println(light);

  if (light < 2000) {
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }

  delay(500);
}
```

Output:

```
Light: 2938
Light: 2938
Light: 2938
Light: 2938
Light: 177
Light: 172
Light: 2965
Light: 4036
Light: 4063
Light: 4063
```

Photos/ Screenshot/Snapshot :



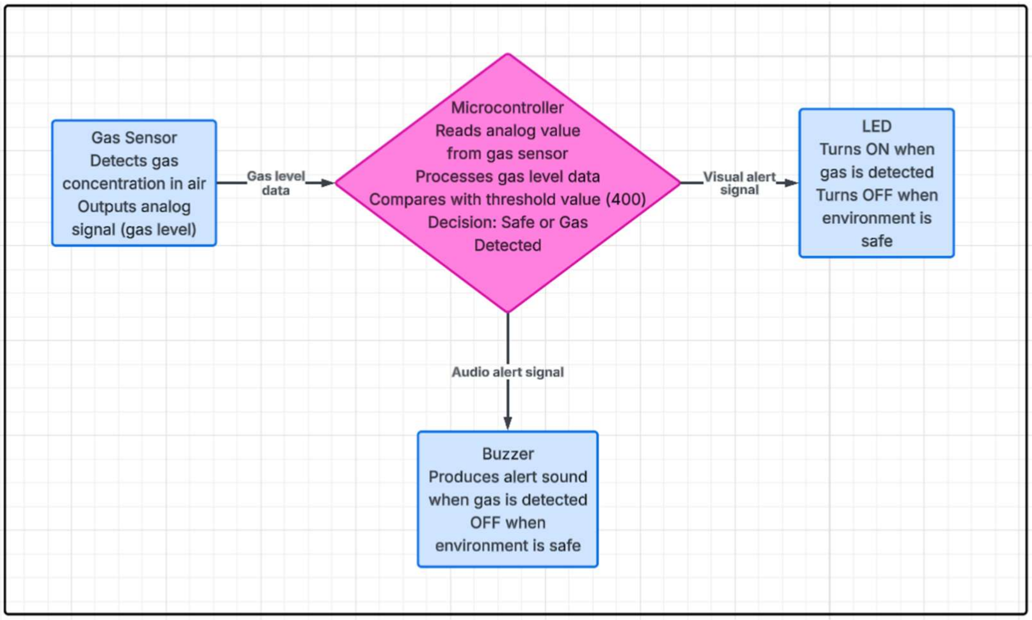
Outcome:

The system successfully detects ambient light intensity and controls the LED accordingly. It demonstrates an automated and energy-efficient lighting solution using IoT concepts.

Activity 3: Wokwi STM

<table>
<tr><td>Problem Statement</td><td colspan="5">Gas leakage poses a serious threat to safety in homes and industrial environments. Traditional detection systems may not provide immediate alerts or continuous monitoring. Therefore, a simple and automated system is required to detect gas levels and provide instant alerts to prevent potential hazards.</td></tr>
<tr><td>AAT MARKS</td><td>Design (5)</td><td>Execution (5)</td><td>Result (5)</td><td>Innovation/ Extra Features (5)</td><td>Total</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>TOOLS USED</td><td colspan="5">• Wokwi
• STM32 Nucleo C031C6
• MQ-2 Gas Sensor
• LED
• Buzzer</td></tr>
<tr><td>MAP SDG</td><td colspan="5">• SDG 3: Good Health and Well-being
Ensures safety by detecting harmful gas levels and providing early alerts.
• SDG 11: Sustainable Cities and Communities
Supports safer living environments through real-time monitoring of gas leakage.
• SDG 9: Industry, Innovation and Infrastructure
Demonstrates the use of embedded systems for developing smart safety solutions.</td></tr>
<tr><td>MAP COURSE OUTCOME of IOT/ BFE</td><td colspan="5"><table><tr><th>CO</th><th>Mapping</th><th>Justification</th></tr><tr><td>CO 1</td><td>Yes</td><td>Demonstrates IoT perception layer using gas sensor to detect gas concentration, with microcontroller processing and generating alerts</td></tr></table></td></tr>
</table>

	CO 2	Yes	Uses Arduino C programming, analog data acquisition, and signal processing for gas detection and alert generation
	CO 3	No	Data analytics is not applied as the system focuses on real-time monitoring.
	CO 4	Yes	Integrates gas sensor (sensing) with LED and buzzer (actuation) to implement a complete sense-process-act IoT system
MAP PROGRAM OUTCOME	**PO**	**Mapping**	**Justification**
	PO 1	Yes	Applied knowledge of sensors, electronics, and programming to detect gas levels and trigger alerts
	PO 2	Yes	Identified and analyzed real-world problems such as gas leakage and safety hazards
	PO 3	Yes	Designed an automated gas detection and alert system for safety applications
	PO 5	Yes	Used modern tools such as Arduino IDE and simulation platforms for implementation
	PO 6	Yes	Addresses environmental and societal concerns by detecting harmful gases and providing alerts
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	**PSO**	**Mapping**	**Justification**
	PSO 1	Yes	Demonstrates programming and logical implementation of gas detection using sensor input and threshold conditions
	PSO 2	Yes	Solves real-world problems like gas leakage detection and safety monitoring

<table>
<tr><td></td><td><table><tr><td>PSO 3</td><td>Yes</td><td>Utilizes simulation tools like Wokwi to design, test, and validate the gas detection system</td></tr></table></td></tr>
<tr><td>URL Link</td><td>https://wokwi.com/projects/461375937572056065</td></tr>
<tr><td>SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model</td><td>
</td></tr>
<tr><td>COURSE COORDINATOR</td><td>Dr ARUNA M G</td></tr>
</table>

Details
Abstract : The Gas Detection System is an IoT-based project developed using STM32 and an MQ-2 gas sensor. The system continuously monitors gas levels and provides alerts when the concentration exceeds a predefined threshold. When gas is detected, an LED glows and a buzzer sounds to indicate danger. This project demonstrates real-time monitoring and safety alert mechanisms using embedded systems.
Use Case:

This system can be used in homes, industries, and gas storage areas to detect gas leakage and provide immediate alerts, ensuring safety and preventing accidents.

Introduction:

Gas leakage is a major safety concern in residential and industrial environments. Early detection is essential to prevent accidents and hazards. This project uses a gas sensor to monitor gas levels and an STM32 microcontroller to process the data and trigger alerts. It demonstrates a simple and effective safety system using embedded technology.

Objective:

- To detect gas concentration using an MQ-2 gas sensor
- To interface the gas sensor with STM32 microcontroller
- To monitor gas levels in real time
- To trigger LED and buzzer alerts when gas exceeds a threshold
- To develop a simple and effective gas leakage detection system

General procedure

- Connect gas sensor to STM32 (VCC, GND, AO)
- Connect LED and buzzer to output pins
- Initialize serial communication
- Read gas sensor values
- Compare with threshold
- Trigger LED and buzzer based on gas level

Step to Execute:

- Open Wokwi simulation platform
- Add STM32, gas sensor, LED, and buzzer
- Connect all components properly
- Paste the code
- Start simulation
- Adjust gas sensor values

- Observe LED, buzzer, and serial output

Source Code

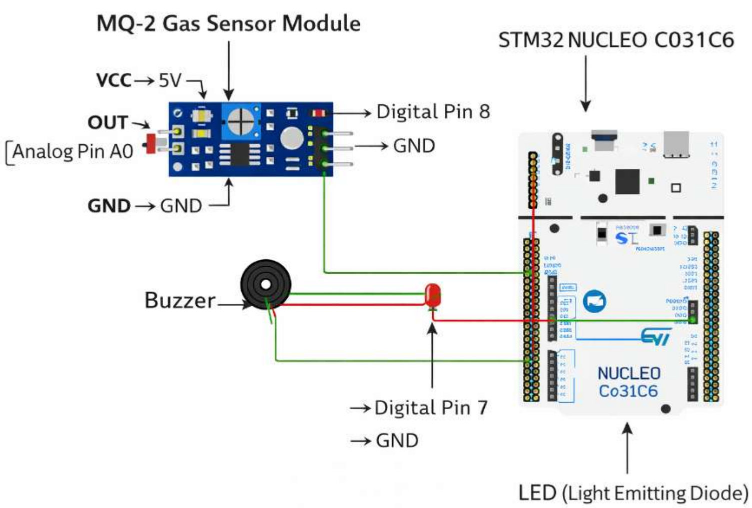
```
int gasPin = A0;

int led = 7;

int buzzer = 8;

void setup() {

  pinMode(led, OUTPUT);

  pinMode(buzzer, OUTPUT);

  Serial.begin(9600);

}

void loop() {

  int gasValue = analogRead(gasPin);


  Serial.print("Gas Level: ");

  Serial.println(gasValue);


  if (gasValue > 400) {

    digitalWrite(led, HIGH);

    digitalWrite(buzzer, HIGH);

    Serial.println("Gas Detected!");

  } else {

    digitalWrite(led, LOW);

    digitalWrite(buzzer, LOW);

    Serial.println("Safe");
```

```
  }



  Serial.println("----------------");

  delay(1000);

}
```

Output:

```
-----------
Gas Level: 892
Gas Detected!
-----------
Gas Level: 562
Gas Detected!
-----------
Gas Level: 255
Safe
-----------
Gas Level: 255
Safe
-----------
```

Photos/ Screenshot/Snapshot :



Outcome:

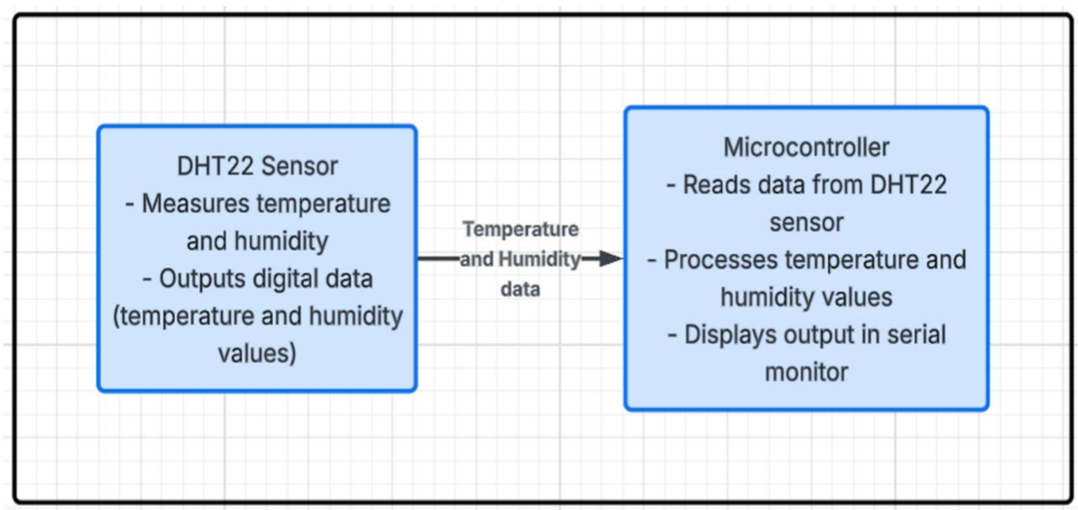
The system successfully detects gas levels and provides alerts using LED and buzzer. It demonstrates a simple and effective safety monitoring system using STM32.

Activity 4: Wokwi RaspberryPi Pico simulator

<table>
<tr><td>Problem Statement</td><td colspan="5">Monitoring temperature conditions is essential in many applications such as environmental monitoring and safety systems. However, raw sensor values alone are not always meaningful for decision-making. Therefore, a system is required to intelligently classify temperature data into meaningful categories to provide better insights and alerts.</td></tr>
<tr><td>AAT MARKS</td><td>Design (5)</td><td>Execution (5)</td><td>Result (5)</td><td>Innovation/ Extra Features (5)</td><td>Total</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>TOOLS USED</td><td colspan="5">• Wokwi
• Raspberry Pi
• DHT22</td></tr>
<tr><td>MAP SDG</td><td colspan="5">• SDG 3: Good Health and Well-being
Helps in monitoring environmental conditions to ensure safe and healthy surroundings.
• SDG 9: Industry, Innovation and Infrastructure
Demonstrates the real-time monitoring using embedded systems.
• SDG 13: Climate Action
Supports monitoring of temperature changes, contributing to environmental awareness.</td></tr>
<tr><td>MAP COURSE OUTCOME of IOT/ BFE</td><td colspan="5"><table><tr><td>CO</td><td>Mapping</td><td>Justification</td></tr><tr><td>CO1</td><td>Yes</td><td>Demonstrates IoT perception layer using DHT22 sensor to measure temperature and humidity, with microcontroller processing the data</td></tr></table></td></tr>
</table>

	CO2	Yes	Uses MicroPython programming, sensor interfacing, and data acquisition from DHT22 sensor
	CO3	No	Data analytics is not included as the system operates in real-time for automatic lighting control without storing or analyzing historical data
	CO4	Yes	Integrates DHT22 sensor (sensing) and microcontroller processing to implement a complete data acquisition system
MAP PROGRAM OUTCOME	**PO**	**Mapping**	**Justification**
	PO1	Yes	Applied knowledge of sensors, environmental parameters, and programming to measure temperature and humidity
	PO2	Yes	Identified and analyzed the need for environmental monitoring in real-world applications
	PO3	Yes	Designed a system to monitor and display temperature and humidity in real time
	PO5	Yes	Used modern tools such as Wokwi and MicroPython for system implementation
	PO6	Yes	Contributes to environmental monitoring and maintaining safe living conditions
MAP PROGRAM SPECIFIC OUTCOME(PSO) of DEPARTMENT	**PSO**	**Mapping**	**Justification**
	PSO 1	Yes	Demonstrates programming and problem-solving by implementing sensor-based logic to acquire environmental data

<table>
<tr><td></td><td><table>
<tr><td>PSO 2</td><td>Yes</td><td>Addresses real-world problems like monitoring temperature and humidity in homes and industries</td></tr>
<tr><td>PSO 3</td><td>Yes</td><td>Utilizes embedded/IoT tools or simulation platforms to design, test, and validate the monitoring system</td></tr>
</table></td></tr>
<tr><td>URL Link</td><td>https://wokwi.com/projects/461385917995395073</td></tr>
<tr><td>SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model</td><td>



</td></tr>
<tr><td>COURSE COORDINATOR</td><td>Dr ARUNA M G</td></tr>
</table>

Details
Abstract : The Temperature and Humidity Monitoring System uses a DHT22 sensor and a microcontroller to measure environmental conditions. The sensor collects temperature and humidity data, which is processed and displayed in the serial monitor. This project demonstrates real-time monitoring using embedded systems and highlights basic IoT concepts.

Use Case:

This system can be used in smart homes, agriculture, and industrial environments to monitor temperature and humidity conditions in real time.

Introduction:

Monitoring temperature and humidity is essential for maintaining comfortable and safe environments. Traditional methods are not efficient for continuous tracking. This project uses a DHT22 sensor to measure environmental conditions and a microcontroller to display the values in real time. It demonstrates a simple and effective monitoring system using embedded technology.

Objective:

- To measure temperature using DHT22 sensor
- To measure humidity using DHT22 sensor
- To interface the sensor with a microcontroller
- To display real-time values in serial monitor
- To develop a simple environmental monitoring system

General procedure

- Connect DHT22 sensor to microcontroller
- Initialize sensor and serial communication
- Read temperature and humidity values
- Display values in serial monitor
- Repeat continuously

Step to Execute:

- Open Wokwi simulation platform
- Add microcontroller and DHT22 sensor
- Connect VCC, GND, and DATA pins
- Paste the code
- Start simulation
- Adjust temperature and humidity

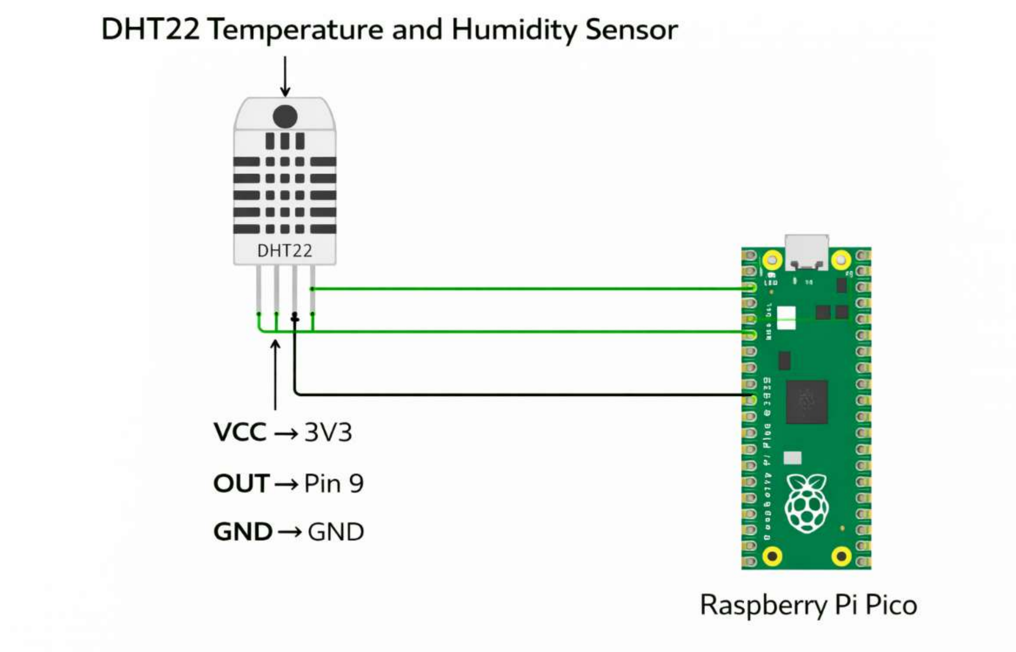
Source Code

```
import dht
from machine import Pin
import time
sensor = dht.DHT22(Pin(15))
while True:
    sensor.measure()
    temp = sensor.temperature()
    hum = sensor.humidity()

    print("Temperature:", temp, "°C")
    print("Humidity:", hum, "%")
    print("----------------")
    time.sleep(2)
```

Output:

```
Temperature: 27.50 °C
Humidity: 58.00 %
----------------

Temperature: 27.80 °C
Humidity: 57.50 %
----------------

Temperature: 28.10 °C
Humidity: 56.90 %
----------------
```

Photos/ Screenshot/Snapshot :



Outcome:

The system successfully measures and displays temperature and humidity in real time using a DHT22 sensor. It demonstrates a simple and effective environmental monitoring system.

Activity 5: TinyML simulation in Wokwi

Problem Statement	Measuring distance alone does not provide meaningful insights for decision-making in real-time systems. There is a need for a system that not only measures distance but also classifies it into actionable categories. This project aims to simulate TinyML by classifying distance data into different levels for better interpretation.

AAT MARKS	**Design (5)**	**Execution (5)**	**Result (5)**	**Innovation/ Extra Features (5)**	**Total**

TOOLS USED	• Wokwi • Arduino Uno • Ultrasonic Sensor (HC-SR04)
MAP SDG	• **SDG 9: Industry, Innovation and Infrastructure** Demonstrates intelligent embedded systems using real-time data processing. • **SDG 11: Sustainable Cities and Communities** Can be used in smart parking and obstacle detection systems. • **SDG 13: Climate Action** Supports development of smart monitoring systems for environmental awareness.

MAP COURSE OUTCOME of IOT

CO	**Mapping**	**Justification**
CO1	Yes	Demonstrates IoT perception layer using ultrasonic sensor to measure distance, with microcontroller processing and classification output.
CO2	Yes	Uses Arduino C programming, sensor interfacing, signal processing (trigger and echo), and decision logic for classification
CO3	No	Applies basic data interpretation by classifying distance into categories (Very Close, Near, Far), simulating data analysis

CO4	Yes	Integrates ultrasonic sensor (sensing) and processing logic to implement a complete sense-process-act system

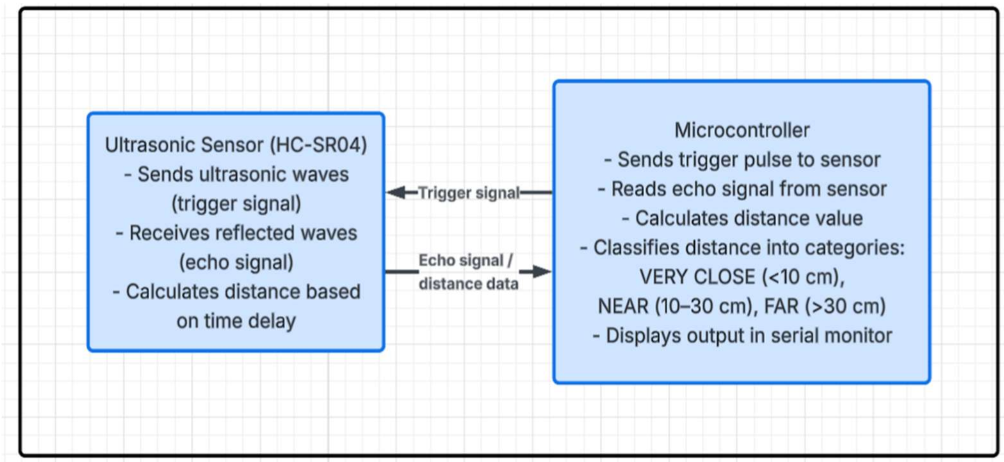
MAP PROGRAM OUTCOME

PO	**Mapping**	**Justification**
PO1	Yes	Applied knowledge of sensors and embedded systems for processing distance data
PO2	Yes	Identified and analysed the limitation of traditional physical switches in modern smart environments
PO3	Yes	Designed a convenient human-machine interaction system to enable contactless or automated control
PO5	Yes	Used modern engineering tools such as microcontrollers and simulation/software platforms for system development
PO6	Yes	Improves user convenience and accessibility, supporting safer and more hygienic interaction in smart environments

MAP PROGRAM SPECIFIC OUTCOME (PSO) of DEPARTMENT

PSO	**Mapping**	**Justification**
PSO 1	Yes	Demonstrates programming and logical implementation of distance measurement and classification using sensor data
PSO 2	Yes	Solves real-world problems such as obstacle detection and distance-based decision systems
PSO 3	Yes	Utilizes embedded systems or IoT simulation tools to design, test, and validate a smart interaction-based control system

URL Link: https://wokwi.com/projects/461389991923891201

SYSTEM DIAGRAM / UML / SEQUENCE DIGRAM/ 3D Model	
COURSE COORDINATOR	Dr ARUNA M G

Details
Abstract : The Ultrasonic Distance Classification System uses an HC-SR04 sensor to measure distance and classify it into different categories. The system processes real-time distance data and categorizes it as very close, near, or far. This classification simulates TinyML inference, where data is analyzed on-device to generate meaningful outputs. The project demonstrates intelligent decision-making in embedded systems.
Use Case: This system can be used in obstacle detection, smart parking systems, and robotics to identify object proximity and take appropriate actions.
Introduction: Distance measurement is widely used in automation and smart systems. However, raw distance values are not always sufficient for decision-making. TinyML enables on-device intelligence by processing data and providing meaningful classifications. This project uses an ultrasonic sensor to measure distance and classify it into predefined categories, simulating TinyML behavior.

Objective:

- To measure distance using an ultrasonic sensor
- To interface the sensor with a microcontroller
- To process real-time distance data
- To simulate TinyML-based inference on embedded systems

General procedure

- Connect ultrasonic sensor to microcontroller
- Initialize pins and serial communication
- Generate trigger pulse
- Measure echo signal
- Calculate distance
- Classify distance into categories
- Display output

Step to Execute:

- Create project in Edge Impulse (optional)
- Upload audio samples (clap and noise)
- Train audio classification model
- Deploy model as Arduino library
- Open Wokwi and select ESP32
- Paste code and run simulation
- Observe output in Serial Monitor

Source Code

```
define TRIG 12

#define ECHO 5

void setup()
```

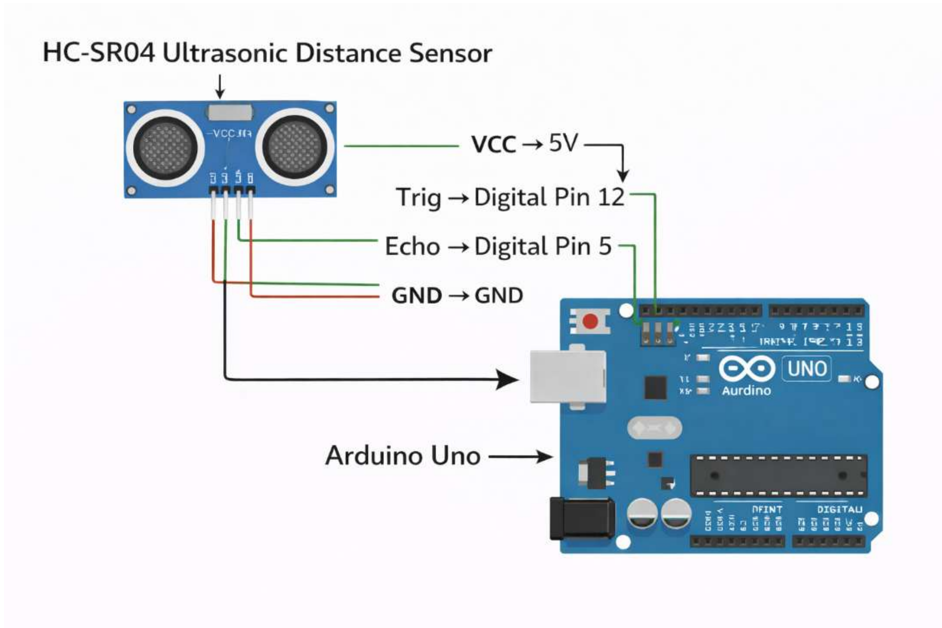
```
{
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  Serial.begin(9600);
}
void loop()
{
  long duration;
  float distance;
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  duration = pulseIn(ECHO, HIGH);
  distance = duration * 0.034 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
  if (distance < 10)
```

```
{
  Serial.println("Class: VERY CLOSE");}
 else if (distance < 30)
{
  Serial.println("Class: NEAR");}
 else { Serial.println("Class: FAR"); }
 Serial.println("----------------------");
 delay(500);
}
```

Output:

```
----------------------
Distance: 279.26 cm
Class: FAR
----------------------
Distance: 307.16 cm
Class: FAR
----------------------
Distance: 337.94 cm
Class: FAR
----------------------
Distance: 2.01 cm
Class: VERY CLOSE
----------------------
Distance: 2.01 cm
Class: VERY CLOSE
----------------------
Distance: 2.01 cm
Class: VERY CLOSE
----------------------
Distance: 2.01 cm
Class: VERY CLOSE
----------------------
Distance: 88.40 cm
Class: FAR
----------------------
Distance: 88.50 cm
Class: FAR
----------------------
```

Photos/ Screenshot/Snapshot :



Outcome:

The system successfully measures and classifies distance into meaningful categories. It demonstrates how embedded systems can perform real-time decision-making, simulating TinyML inference.

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